Engineering Structures 262 (2022) 114325

Contents lists available at ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct

Probabilistic vortex-induced vibration occurrence prediction of the twin-box girder for long-span cable-stayed bridges based on wind tunnel tests

Baixue Ge, Rujin Ma*, Fangkuan Li, Xiaohong Hu, Airong Chen

College of Civil Engineering, Tongji University, 200092 Shanghai, China

ARTICLE INFO

Keywords:
Separated twin-box girder
Vortex-induced vibration
Structural damping ratio
Wind speed
Wind direction
Decision tree
Aerodynamic countermeasures
Cost-benefit

ABSTRACT

The twin-box girder is a recommendable structural style for long-span cable-supported bridges due to its excellent aeroelastic stability. Vortex-induced vibration (VIV) is a typical wind-induced vibration for this kind of girder with a high occurrence probability associated with wind characteristics, structural properties and aerodynamic configuration under uncertainty. Therefore, probabilistic approaches should be adopted to predict the occurrence of VIV under uncertainty. This paper presents such an approach to predict the VIV occurrence using the decision tree method. Wind tunnel tests on a large-scale section model are conducted to investigate the effects of structural damping ratio and wind speed on VIV performance. Uniform and turbulent wind flow fields under three attack angles of -3°, 0°, and -3° are tested. Moreover, wind direction at the bridge site is determined as a parameter affecting VIV occurrence. As a result, a decision tree considering the effects of structural damping ratio, wind direction and wind speed is formulated to estimate the probability of VIV occurrence. In order to suppress VIV, aerodynamic countermeasures (ACMs) with L-shaped windshields are applied. Furthermore, the probabilistic cost-benefit analysis of the application of ACMs is developed using a decision tree, and the probability of the positive cost-benefit from ACMs application is accordingly estimated. The application of the proposed probabilistic approach is investigated on a long-span cable-stayed bridge with two separated steel boxes.

1. Introduction

Existing bridges with long spans and low weight are highly sensitive to wind actions due to their extremely flexible and slight inherent damping [33,10]. During recent decades, twin-box girders were proposed and adopted for several long-span bridges due to its superior aerodynamic stability. However, as a separated structure, the twin-box girder is usually faced with an inevitable issue of wind-induced phenomena which will decrease structural functionality and serviceability [14,48,57]. Among various wind-induced vibrations, vortex-induced vibration (VIV) is considered as a frequently occurred and serious aerodynamic phenomenon [35,50]. During the last few decades, representative VIV events have been found in many real bridges, such as Kessock Bridge in Scotland [40], Great Belt East Bridge in Denmark [13], Trans-Tokyo Bay Crossing Bridge in Japan [14], Xihoumen Bridge in China [32] and Humen Bridge in China [16]. Although the VIV of long-span bridges is a limited amplitude vibration and rarely leads to the catastrophic collapse of bridges, it will significantly affect the driving comfort of users and cause long-term fatigue damages due to the high occurrence probability under the low wind velocity conditions [7]. Moreover, the occurrence of VIV is highly affected by the uncertainties associated with the structural properties, wind characteristics, geometric parameters, and aerodynamic configuration [34,62]. Accordingly, a probabilistic approach for accurate VIV occurrence prediction is urgently required for effectively and efficiently risk and lifetime management of long-span bridges.

Wind tunnel test is a direct and powerful method to investigate the VIV of long-span bridges prior to the design or/and completed constructions. Generally, large-scale section models with similar modal dynamics (i.e., modal mass, modal frequency, and modal damping) and configurations with the prototype bridges are used for the wind tunnel test. [52]. Zhou & Ge [58] experimentally studied the vehicle's effects on VIV performance of the Shanghai Bridge over Yangtse River using a 1:60 scale section model of the deck. Trush et al. [46] conducted wind-

* Corresponding author.
E-mail addresses: baixuege@tongji.edu.cn (B. Ge), rjma@tongji.edu.cn (R. Ma), 1lifangkuan@tongji.edu.cn (F. Li), xhu@tongji.edu.cn (X. Hu), a.chen@tongji.edu.cn (A. Chen).

https://doi.org/10.1016/j.engstruct.2022.114325
Received 11 November 2021; Received in revised form 12 April 2022; Accepted 24 April 2022
Available online 7 May 2022

tunnel experiments on the VIV of rough bridge cables, particularly focusing on the effects of surface roughness and atmospheric turbulence. Optimum aerodynamic countermeasures (ACMs) for VIV suppression of a super-wide streamline box girder were experimentally introduced by Li et al. [35]. The ACMs achieved by adding additional aerodynamic components to bridge girders are one of the commonly used approaches. The popular aerodynamic components involved in ACMs include wind fairings, guide vanes, spoilers, and wind barriers. On the other hand, the modification of airflow around the bridge girder by installing or adjusting the parameters of some non-structural components with particular shape (e.g. railings, maintenance rail, crash barriers) at specific location of the girder has also proved to be useful [53,54,59]. Experimental investigations on the characteristics of bridge's VIV and its ACMs were presented in Liu et al. [36]. Despite the fact that the VIV of long-span bridges has received a lot of research attention, it cannot be completely avoided during the lifetime. As a result, productive VIV control strategies should be further explored and implemented.

Researchers, through time, have proposed a series of effective countermeasures to mitigate bridge VIV of long-span bridges. These countermeasures are typically divided into three types: aerodynamic [35,42,56,63], structural and mechanical [45]. Because of the advantages of simple installation, easy maintenance and high efficiency, ACMs are often used for the VIV suppression of bridges [1]. Over the last few decades, extensive applications of ACMs on long-span bridges for VIV mitigation and associated investigations have been developed. El-Gammal et al. [12] carried out an experimental study on the determination of the effectiveness of geometric spanwise sinusoidal perturbation in controlling the VIV of a plate girder bridge. The influences of handrail, inspection vehicle rail and guide vane on the VIV of a closed box girder were discussed in Sun et al. [44]. Laima et al. [28] introduced the effects of attachments in terms of handrails, crash barriers, wind barriers and maintenance rails on VIV of a separated twin-box girder with wind tunnel tests. The aforementioned studies are available for (a) understanding the effects of various ACMs on VIV of multiple types of long-span bridges; (b) exploring the optimum and effective VIV control measures to minimize VIV responses. However, little investigation focuses on the economic investments of ACMs application against their effectiveness and benefits. Cost-benefit analysis is an extensively applied and reliable method supporting risk mitigation measures such as ACMs considering the effects of costs and benefits simultaneously. To the authors' best knowledge, there are no studies available for the cost-benefit analysis of VIV application. Therefore, it is necessary to develop the cost-benefit analysis from VIV application for a more reasonable manager's decision-making.

Considering the multiple parameters that can affect VIV occurrence under uncertainty, this paper proposes novel probabilistic approaches for predicting VIV occurrence using the decision tree method which is widely used for classification tasks. One of the advantages of using the decision tree over other methods is that this approach does not require any input pre-processing such as data normalization, scaling or centering and can handle data on different measurement scales. In addition, such a tree represents a good compromise between comprehensibility, accuracy, and efficiency. Moreover, wind tunnel tests are conducted to investigate the effects of structural damping ratio and wind speed on VIV performance. To understand the influences of turbulence intensity on VIV, uniform and turbulent wind flow fields under three wind attack angles are tested. Furthermore, wind direction is determined as an important parameter affecting VIV occurrence. Finally, a decision tree is formulated to predict the probability of VIV occurrence, independently considering the probabilities of structural damping ratios, wind direction and wind speed for VIV occurrence. The uncertainties of structural damping ratios and wind speed are addressed to predict VIV using appropriate random variables and best-fitted distributions. Moreover, ACMs with L-shaped windshields are experimentally studied in terms of their effects on VIV performance. Cost-benefit analysis from ACMs application using a decision tree is further described. Considering the uncertainties of the costs related to the ACMs, the probability that ACMs lead to the positive cost-benefit is accordingly estimated. A long-span cable-stayed bridge with a separated twin-box girder is used for the investigations of the presented study.

2. Vortex-induced vibration

VIV is one of the representative wind-induced vibrations for long-span bridges, resulting in long-term fatigue damages and decreasing structural safety and functionality [18,19,31,42]. Fig. 1(a) shows the relation of vortex shedding frequency and the natural frequency of the bridge. As illustrated in Fig. 1(a), with increasing wind speed U, the vortex shedding frequency f_v increases initially, and subsequently, when the vortex shedding frequency f_v is identical to the natural frequency of the bridge f_s within a lock-in wind speed range, VIV occurs. Therefore, the occurrence of VIV corresponds to lock-in wind speed intervals which can be measured by the wind tunnel test and structural health monitoring (SHM) systems [2]. In this study, the wind tunnel test is conducted to capture the lock-in wind speed interval of VIV.

When the bridge suffers from VIV during a lock-in wind speed range, it will lead to the large displacement of bridge section, as shown in Fig. 1 (b). To ensure bridge safety, usability and driving comfort, VIV responses should be controlled below an allowable level. According to the Chinese bridge wind-resistant specification [38], the vertical and torsional allowable VIV amplitudes can be calculated, respectively, as.

$$h_{v,lim} = \gamma_v \cdot \frac{0.04}{f_b} \tag{1a}$$

$$\theta_{\alpha,lim} = \gamma_v \cdot \frac{4.56}{Bf_\alpha} \tag{1b}$$

where $h_{v,lim}$ = vertical VIV allowable amplitude (m); f_b = first vertical

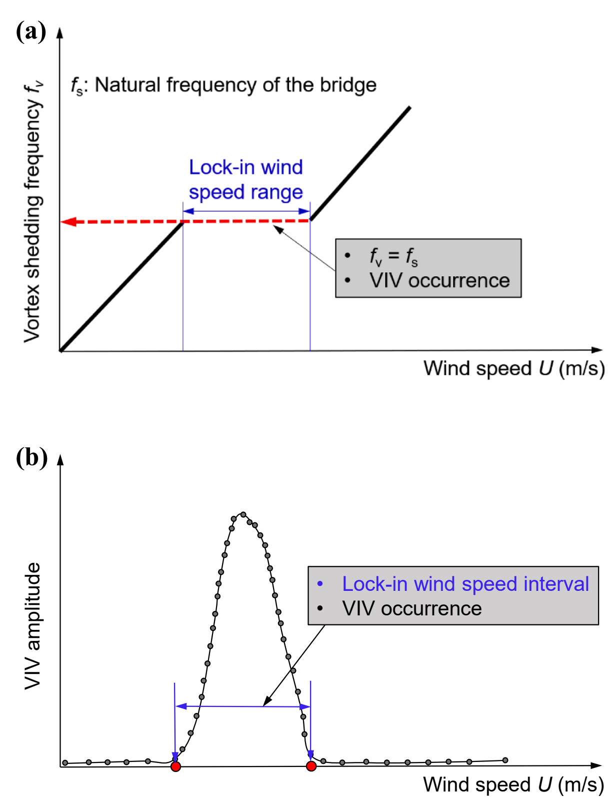

Fig. 1. (a) Relation of the vortex shedding frequency and bridge natural frequency; (b) relation of VIV amplitude and wind speed; (adapted from Williamson & Govardhan [49]).

bending mode frequency (Hz); $\theta_{\alpha,lim}$ = torsional VIV allowable amplitude (°); B = width of the main girder (m); f_α = first torsional mode frequency (Hz). γ_v is the partial coefficient of VIV and is set to be 1.0 when the VIV response is obtained from wind tunnel tests [38]. It should be noted that the values of all parameters in Eq. (1) correspond to those of the prototype bridge.

3. Experimental arrangements

To investigate the performance of VIV and determine the parameters affecting VIV occurrence, in this study, the wind tunnel test which is treated as one of the most accurate and effective ways for capturing the wind-induced effects on bridges, is conducted.

3.1. Description of the investigated bridge

The proposed probabilistic approach to predict VIV occurrence and estimate the cost-benefit from ACMs application is investigated with a symmetrical long-span cable-stayed bridge, Puyi Highway Jiajiang Bridge, located in Nanjing, China. The bridge's construction was completed and it opened to traffic in 2020. The main span of this bridge is 500 m, as presented in Fig. 2(a). The bridge deck is composed of separated steel box girders with a total width of 54.40 m and a depth of 4.0 m, as shown in Fig. 2(b).

3.2. Set-up of the wind tunnel tests

In order to ensure safety and serviceability, wind tunnel tests of the Puyi Highway Jiajiang Bridge have been conducted at the design stage. The wind tunnel test is based on a large-scale section model of the main girder. The section model wind tunnel tests were carried out in the TJ-3 boundary layer wind tunnel at Tongji University. The TJ-3 boundary layer wind tunnel is a vertically arranged low-speed closed-circuit wind tunnel with a test section that measures 14.0 m, 15.0 m and 2.0 m in length, width and height, respectively. The wind speed in this wind tunnel can be continuously adjusted from 1.0 m/s to 17.60 m/s. Other detailed information on the TJ-3 boundary layer wind tunnel can be found in Chen et al. [3].

Based on the cross-section size of the main girder and the test section, 1:30 is chosen as the geometrical scale of the section model. Fig. 3(a) shows the photograph of the large scale section model in the wind tunnel. The main test parameters of wind tunnel tests are listed in Table 1. The length, width, and height of this section model are 3.60 m, 1.813 m and 0.133 m, respectively. This section model is composed of separated steel box girders with a gap width of 0.343 m. The mass per unit length and mass moment of inertia per unit length are 33.96 kg/m and 10.79 kg·m^2/m, respectively. The vertical natural frequency and damping ratio of this section model are 3.31 Hz and 0.13%, respectively, and the torsional natural frequency and damping ratio are, respectively, with the values of 8.77 Hz and 0.25%. Fig. 3(b) illustrates the barriers and handrail configuration adopted in this section model. As shown in this figure, steel handrails and acrylonitrile–butadienestyrene barriers are used.

The free vibration test of the spring-supported (eight springs) section model is used to acquire the aerodynamic phenomenon such as wind-induced responses of this section model. The VIV is sensitive to the structural damping ratio. In order to adjust and simulate the vertical and torsional damping ratios of the system, eight circular energy-dissipation damping rings with adjustable size and section are attached on the eight springs of the section model. However, this method is not suitable to adjust the damping ratios of vertical and torsional motions independently. Furthermore, four laser displacement sensors installed on the end wall of the section model, as presented in Fig. 3(c), are employed to measure the dynamic responses of the section model. A sampling frequency of 256 Hz is used for the displacement signal collecting.

4. Performance of vortex-induced vibration

The displacements induced by VIV are measured using four laser displacement sensors, as described in the previous section, and converted to those of the prototype bridge using the predefined geometric scale ratio. Generally, prototype bridges are immersed in turbulent wind flow fields. Therefore, wind turbulence characteristics may affect VIV performance [5,9,51,57]. To investigate the influence of turbulence

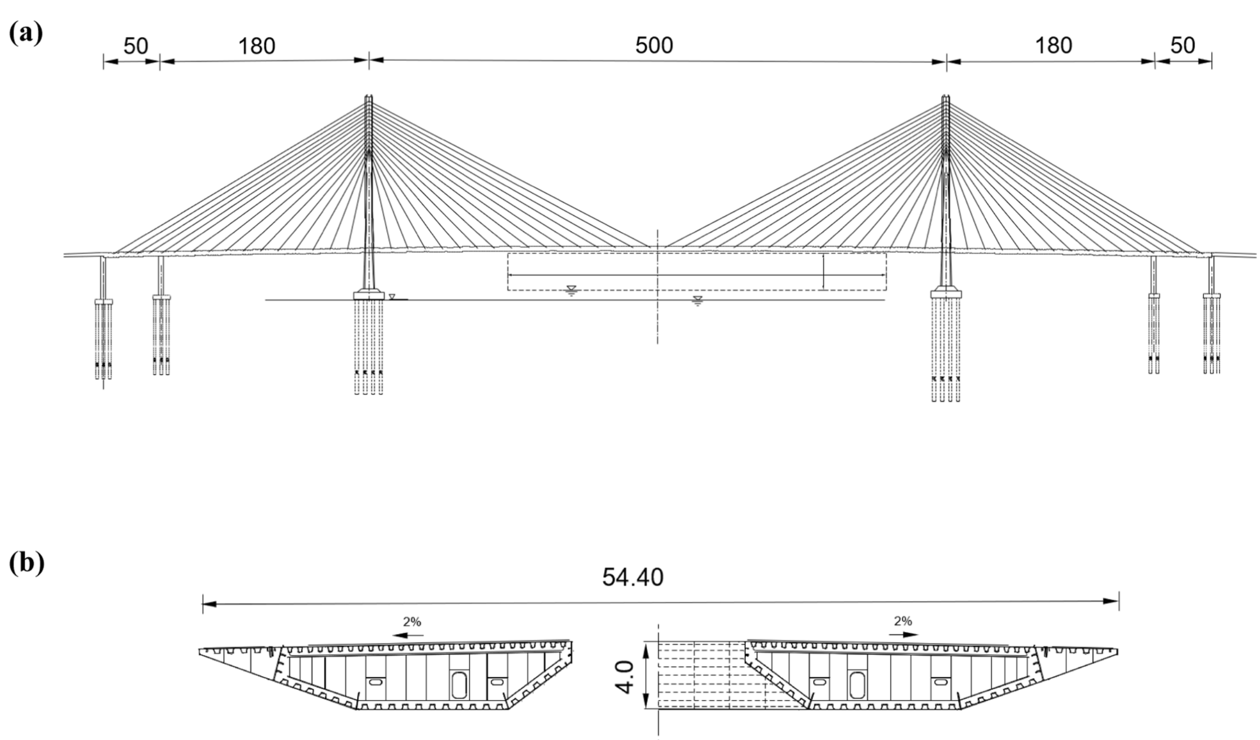

Fig. 2. Puyi Highway Jiajiang Bridge: (a) elevation of bridge; (b) two separated steel box girders; (Unit. m).

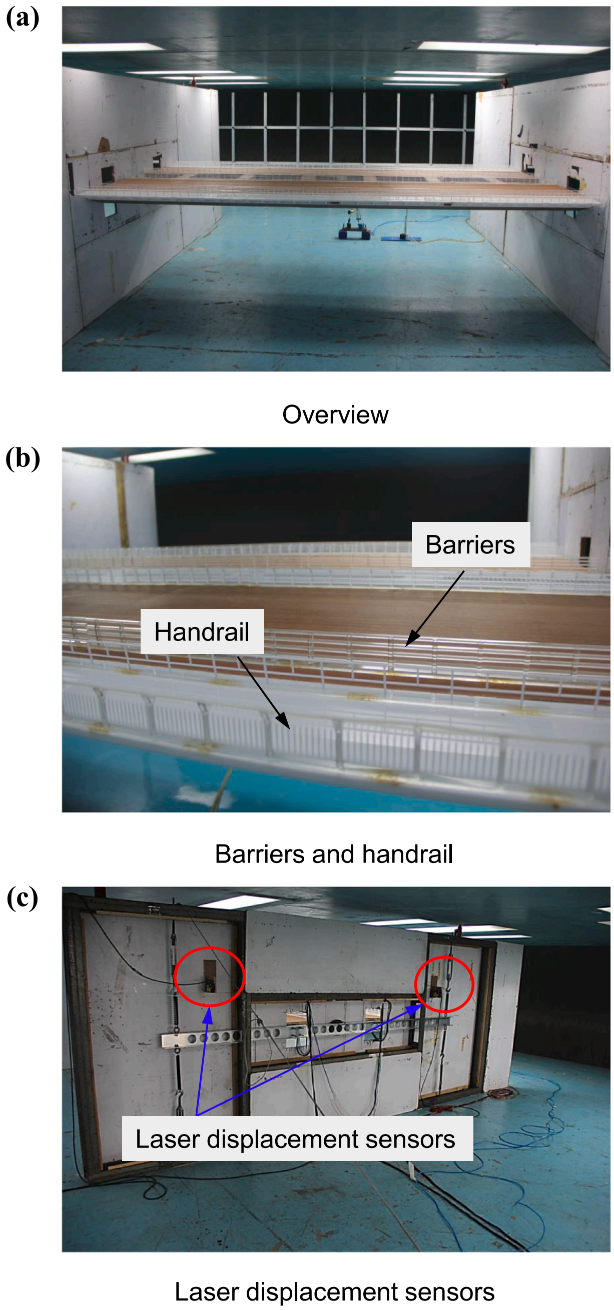

Fig. 3. Section model and measurement system in the wind tunnel test: (a) overview; (b) barriers and handrail; (c) laser displacement sensors.

Table 1
Parameters of large-scale section model wind tunnel tests.

Parameters	Unit	Scale	Prototype bridge	Section model
Length (L)	m	1:30	–	3.60
Height (H)	m	1:30	4.0	0.133
Width (B)	m	1:30	54.40	1.813
Mass per unit length (m)	kg/m	$1{:}30^2$	30,560	33.96
Mass moment of inertia per unit length (I_m)	$kg{\cdot}m^2/m$	$1{:}30^4$	8,738,000	10.79
Vertical frequency f_b	Hz	–	0.315	3.31
Torsional frequency f_α	Hz	–	0.835	8.77

intensity on VIV performance, two cases of wind flow fields are considered as follows:

- Case 1: Uniform wind flow field.
- Case 2: Turbulent wind flow field.

The uniform wind flow field in Case 1 indicates that the turbulence intensity of wind flow fields is close to zero, which is less than that of Case 2, where the turbulent flow field with a turbulence intensity of 2.80% in the along-wind direction is used. In this study, the turbulent flow instrumentation (TFI) dynamic cobra probe measures the three components of wind speed and turbulence characteristics.

The VIV responses of the section model are firstly tested at three different wind attack angles of $+3^\circ$, 0°, and -3° for Cases 1 and 2, where low damping ratios of 0.13% and 0.25% are considered, respectively. Based on the Chinese bridge wind-resistant specification [38], three wind attack angles (i.e., $+3^\circ$, 0°, and -3°) are selected. Fig. 4(a) and (b) plot vertical and torsional VIV amplitudes with increased wind speed at a low damping ratio of 0.13% in Case 1. If Case 2 is considered, the tested results (i.e., vertical and torsional VIV amplitudes) at a low damping ratio which is 0.25% are shown in Fig. 5. The VIV amplitudes and wind speeds are both converted to those of the prototype bridge. Based on Eq. (1), the allowable vertical and torsional amplitudes of the investigated bridge are computed as 126.98 mm and 0.10° (i.e., $h_{v,lim}$ = 126.98 mm and $\theta_{\alpha,lim} = 0.10^\circ$). The testing results indicate that significant vertical VIV occurs at three different attack angles in Cases 1 and 2 (see Figs. 4(a) and 5(a)). Furthermore, the vertical VIV responses of the main girder at three wind attack angles in Case 1 and at $+3^\circ$ and -3° attack angles in Case 2 cannot satisfy the allowable amplitude. However, there is no obvious torsional VIV at three attack angles in Case 2 as shown in Fig. 5(b). Fig. 4(b) shows that the torsional VIV occurs at the attack angles of $+3^\circ$ and 0° in Case 1, where VIV responses are much less than the allowable amplitude. Therefore, the vertical VIV should be

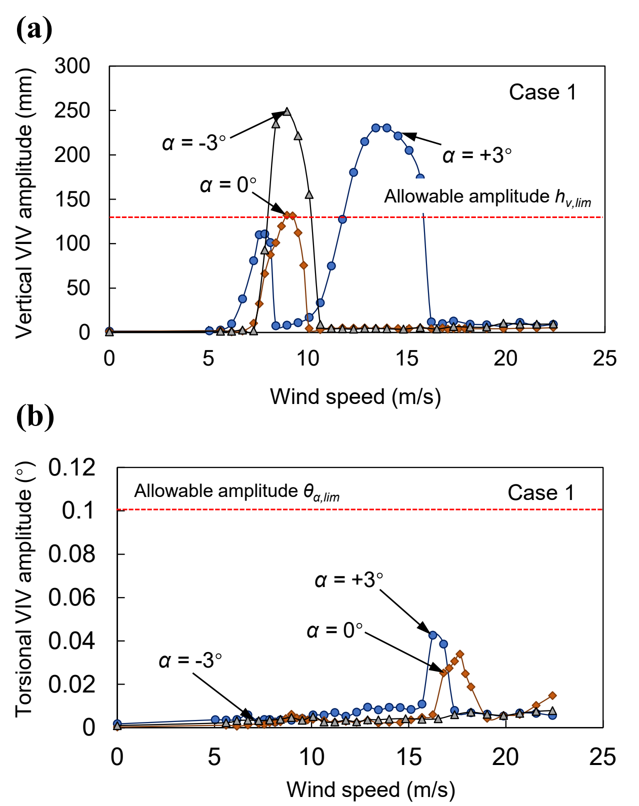

Fig. 4. VIV response of the bridge girder at a low damping ratio in Case 1: (a) vertical; (b) torsional.

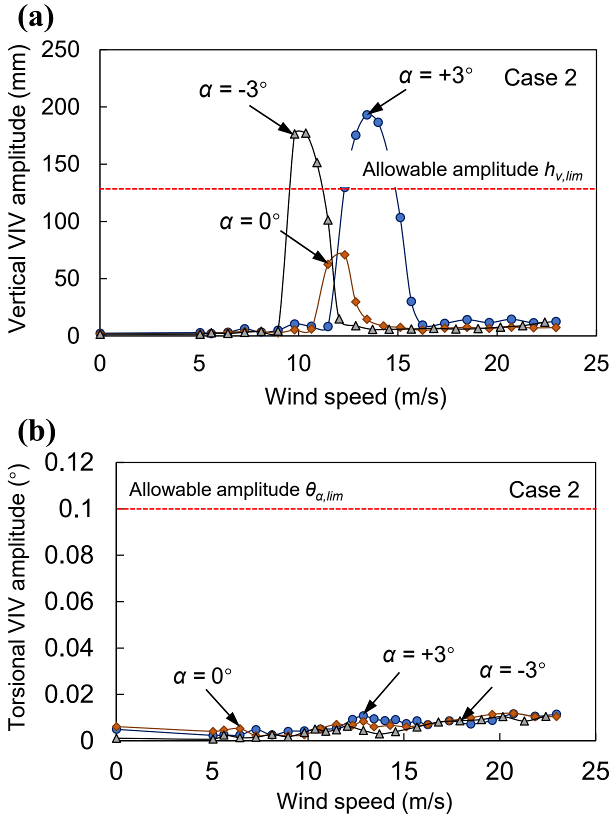

Fig. 5. VIV response of the bridge girder at a low damping ratio in Case 2: (a) vertical; (b) torsional.

further investigated in detail, whereas the torsional VIV of the main girder could be ignored in this study. The detailed descriptions in the following sections are based on the vertical VIV of the main girder for the investigated bridge.

4.1. Effect of structural damping ratio

There is no generally accepted method for estimating the structural damping ratio of bridges before construction due to the considerable complexity and uncertainty. According to the Chinese bridge wind-resistant specification [38], the suggested damping ratio for the highway bridge with steel box girders is 0.3%. To fully investigate the effect of the structural damping ratio ξ on VIV performance, in Case 1 (i.e., uniform wind flow field), structural damping ratios of 0.13%, 0.25%, 0.50%, 0.90%, 1.80% are tested. If Case 2 (i.e., turbulent wind flow field) is used for the wind tunnel test, three damping ratios (i.e., 0.25%, 0.50%, 1.20%) are considered. Fig. 6 shows the vertical VIV amplitude of the bridge girder with increasing wind speed for multiple damping ratios in Case 1. The vertical VIV amplitudes associated with the wind attack angles of +3°, 0°, and −3° in Case 1 are presented in Fig. 6(a), (b) and (c), respectively. Fig. 7 compares the maximum vertical VIV amplitude for multiple damping ratios under the wind attack angles of +3°, 0°, and −3° in Case 1. The maximum VIV amplitudes in Fig. 7 are obtained based on the values provided in Fig. 6. It should be noted that the maximum vertical VIV amplitude is assumed to be zero without VIV occurrence. When the turbulent wind flow field (i.e., Case 2) is tested, the associated vertical and maximum vertical VIV amplitudes of the bridge girder for different damping ratios at three wind attack angles are illustrated in Figs. 8 and 9, respectively.

From Figs. 6, 7, 8 and 9, the following can be seen:

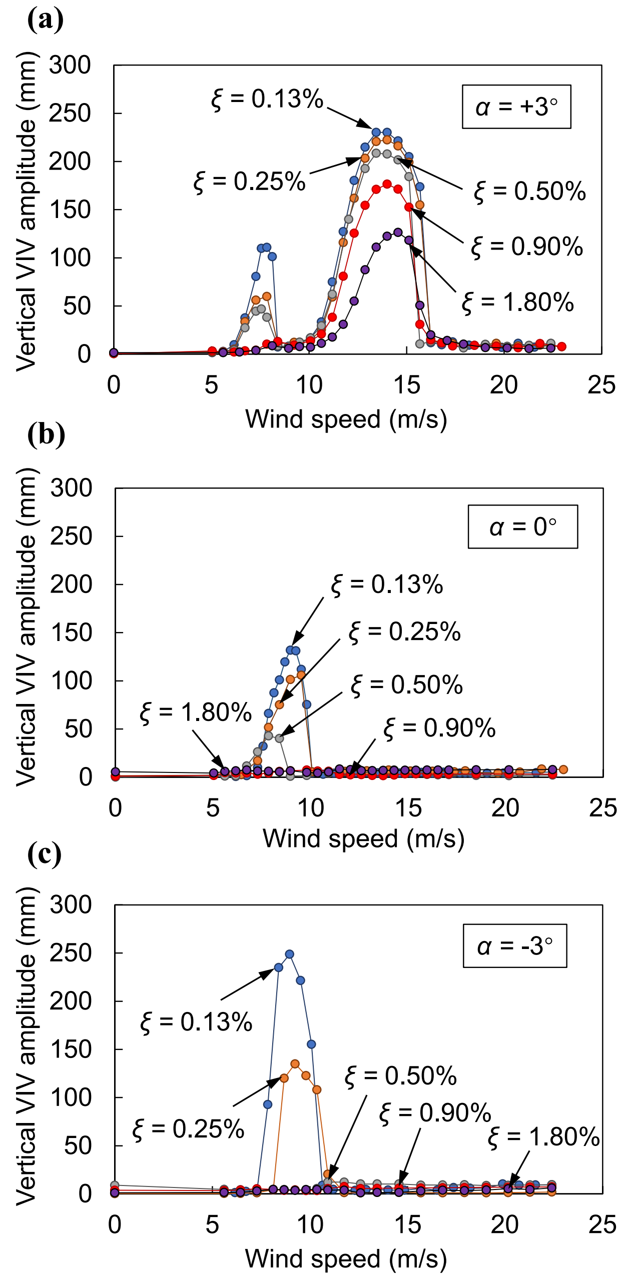

Fig. 6. Vertical VIV amplitude of the bridge girder for multiple damping ratios in Case 1: (a) +3° wind attack angle; (b) 0° wind attack angle; (a) −3° wind attack angle.

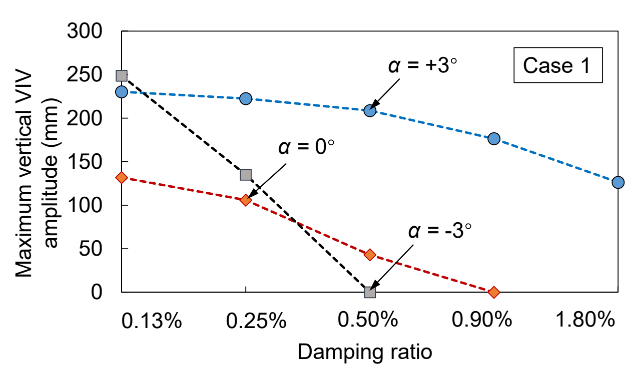

Fig. 7. Maximum vertical VIV amplitude for multiple damping ratios in Case 1.

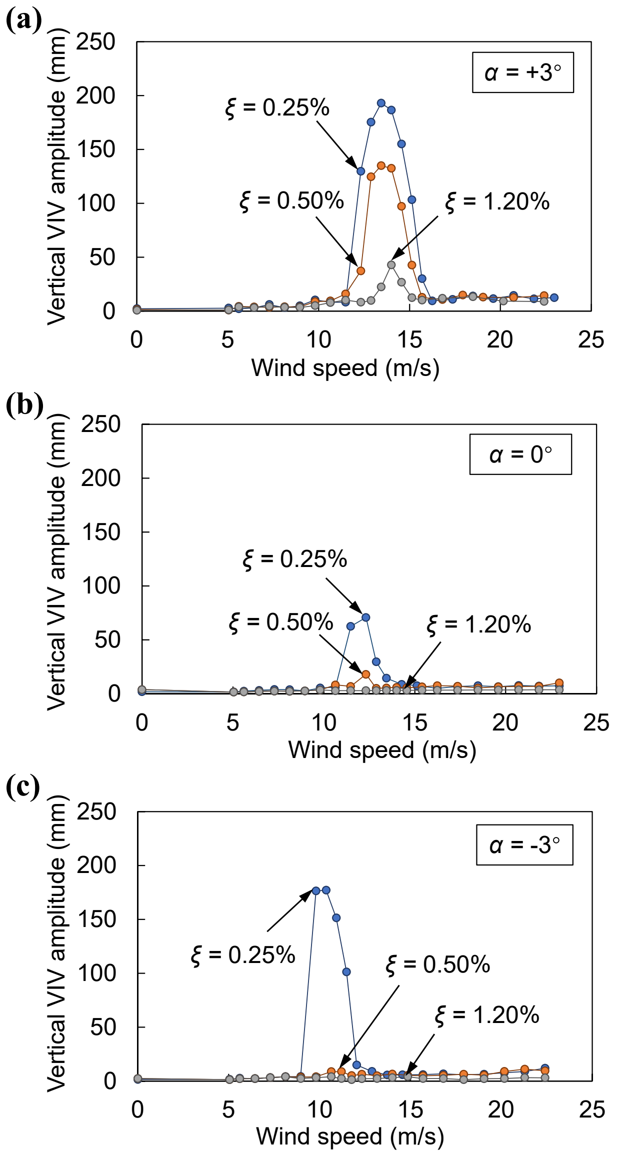

Fig. 8. Vertical VIV amplitude of the bridge girder for multiple damping ratios in Case 2. (a) $+3^\circ$ wind attack angle; (b) 0° wind attack angle; (a) -3° wind attack angle.

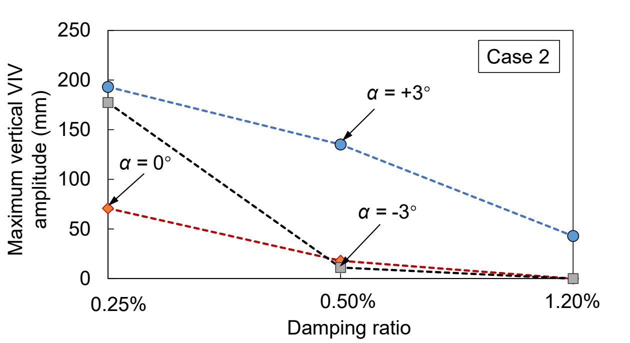

Fig. 9. Maximum vertical VIV amplitude for multiple damping ratios in Case 2.

Under a specific wind attack angle, the vertical and maximum vertical VIV amplitudes decrease as the structural damping ratio ξ increases in both Case 1 (i.e., uniform wind flow field) and Case 2 (turbulent wind flow field).

In Case 1, VIV occurs for all tested damping ratios (i.e., 0.13%, 0.25%, 0.50%, 0.90%, 1.80%) under the wind attack angle of $+3^\circ$. Herein, when the damping ratio reaches 0.90%, the first lock-in wind speed interval of VIV completely disappears (see Fig. 6(a)). If the wind attack angle is tested with 0°, the structural damping ratios of 0.90% and 1.80% have no effect on VIV performance since VIV does not occur (see Fig. 6(b)). Under the wind attack angle of -3°, there is no VIV occurrence for the structural damping ratios of 0.50%, 0.90% and 1.80% (see Fig. 6(c)).

In Case 1, the structural damping of 0.13% leads to the maximum vertical VIV amplitudes of 230.27 mm, 131.87 mm and 248 79 mm under the wind attack angles of $+3^\circ$, 0°, and -3°, respectively (see Fig. 7). All of these maximum values are larger than the vertical allowable amplitude $h_{v,lim} = 126.98$ mm. When the damping ratio increases to 1.80% under the wind attack angle of $+3^\circ$, the maximum vertical VIV amplitude decreases to 126.33 mm, which is less than the $h_{v,lim}$. If the structural damping ratio is 1.80% for the wind attack angles of 0° and -3°, the maximum vertical VIV amplitudes become zero due to no VIV occurrence.

In Case 2, if the structural damping ratio of 1.20% is tested, there will be no VIV occurrence under wind attack angles of 0° and -3° (see Fig. 8(b) and (c), respectively). However, when the structural damping ratio decreases to 0.25%, VIV will occur at three wind attack angles (i.e., $+3^\circ$, 0°, and -3°) as shown in Fig. 8(a), (b) and (c), respectively. For a given wind attack angle, the maximum vertical VIV amplitude for the structural damping of 0.25% exceeds those values associated with the large structural damping ratios (i.e., 0.50% and 1.20%) (see Fig. 9). For example, under the wind attack angle of $+3^\circ$, when the structural damping ratio is tested as 0.25%, the maximum vertical amplitude is measured as 193.02 mm, which is larger than 135.07 mm and 42.92 mm for damping ratios of 0.50% and 1.20%, respectively.

Accordingly, the structural damping ratio can be determined as an important parameter affecting VIV performance. As a result, integrating all tested wind attack angles (i.e., $+3^\circ$, 0°, and -3°), 1.80% and 1.20% are determined as upper bounds of structural damping ratio for VIV occurrence in Case 1 and Case 2, respectively. In other words, VIV will not occur or is in an acceptable level (i.e., allowable amplitudes) when the structural damping ratio is larger than 1.80% and 1.20% of this bridge girder in Case 1 and Case 2, respectively. Moreover, the uncertainty of the structural damping ratio is addressed to predict VIV occurrence by treating it as a random variable.

4.2. Effect of wind speed

Table 2 lists the lock-in wind speed intervals of VIV occurrence in Case 1 and Case 2 at the bridge site. The values shown in Table 2 are based on the measured wind speed intervals of VIV from the wind tunnel test.

From Table 2, the following can be summarized:

VIV occurs only in low average wind speed intervals. Two lock-in wind speed intervals (i.e., the first interval and second interval, denoted as $U_{int,1}$ and $U_{int,2}$, respectively) may cause VIV for the investigated bridge girder. Shiraishi and Matsumoto [43] claimed that response characteristics of VIV were classified into three groups, and the VIV of a majority of bridge sections with wind attack angles between $+7^\circ$ to -7° belongs to the second group, where the vertical oscillation may occur at multiple lock-in wind speed intervals.

In Case 1, VIV occurs during two different lock-in wind speed intervals for the structural damping ratios of 0.13%, 0.25% and 0.50% at the wind attack angle of $+3^\circ$. If the wind attack angles of 0° and -3° are considered in Case 1, only the first wind speed interval $U_{int,1}$ leads to VIV. In Case 2, VIV will only occur within the second wind speed interval $U_{int,2}$ for all tested structural damping ratios.

Table 2
Lock-in wind speed intervals of VIV in Case 1 and Case 2 at the bridge site.

Cases	Damping ratios ξ	Wind attack angle $\alpha = +3^\circ$ Lock-in wind speed interval U_{int} (m/s)		Wind attack angle $\alpha = 0^\circ$ Lock-in wind speed interval U_{int} (m/s)		Wind attack angle $\alpha = -3^\circ$ Lock-in wind speed interval U_{int} (m/s)	
		First interval $U_{int,1}$	Second interval $U_{int,2}$	First interval $U_{int,1}$	Second interval $U_{int,2}$	First interval $U_{int,1}$	Second interval $U_{int,2}$
1	0.13%	6.16–8.40	10.64–16.24	7.28–10.08	–	7.84–10.64	–
	0.25%	6.16–8.40	10.08–16.24	7.28–10.08	–	8.12–10.92	–
	0.50%	6.72–8.40	9.52–15.68	6.72–8.96	–	–	–
	0.90%	–	9.24–15.68	–	–	–	–
	1.80%	–	10.08–16.24	–	–	–	–
2	0.25%	–	11.48–16.24	–	10.36–12.88	–	8.96–12.04
	0.50%	–	11.48–15.68	–	10.64–12.88	–	9.80–11.76
	1.20%	–	12.32–15.12	–	–	–	–

According to Shiraishi and Matsumoto [43], the emergence of multi-"lock-in" regions resulted from the arrival of separated vortices from the leading edge to the trailing edge after n cycles of the heaving motion (where n is a natural number) and the onset critical reduced wind velocity for the smaller-peak oscillation of the vertical mode can be estimated as V_{cr}/n where V_{cr} was the onset critical reduced wind velocity for the larger-peak oscillation.

Considering all structural damping ratios (i.e., 0.13%, 0.25%, 0.50%, 0.90%, 1.80%) under the wind attack angle of $+3^\circ$ in Case 1, the accumulative range of wind speed of VIV is from 6.16 m/s to 16.24 m/s. When the wind attack angle of 0° is tested in Case 1, the wind speed, which ranges from 6.72 m/s to 10.08 m/s, will cause VIV. Moreover, VIV occurs during the wind speed interval of 7.84 m/s to 10.92 m/s, where the wind attack angle of $-3^\circ$is considered in Case 1.

In Case 2, the wind speed resulting in VIV is from 11.48 m/s to 16.24 m/s, 10.36 m/s to 12.88 m/s and 8.96 m/s to 12.04 m/s under the wind attack angles of $+3^\circ$, 0°, and -3°, respectively, where all the tested damping ratios (i.e., 0.25%, 0.50%, 1.20%) are taken into account.

According to the above findings, the wind speed can be treated as an important parameter for the prediction of VIV occurrence. Consequently, the accumulative ranges of wind speed from 6.16 m/s to 16.24 m/s and 8.96 m/s to 16.24 m/s will lead to VIV in Case 1 and Case 2, respectively, where all the tested wind attack angles (i.e., $+3^\circ$, 0°, and -3°) are considered.

5. Probability of vortex-induced vibration occurrence prediction using decision tree

VIV occurs highly depending on the structural properties and wind characteristics under uncertainty [34]. For this reason, VIV occurrence prediction needs to be based on probabilistic approaches. The structural damping ratio and wind speed are determined as two important parameters for predicting VIV occurrence according to the pre-described results from the wind tunnel test. Furthermore, wind direction also requires to be considered in the prediction of VIV occurrence since VIV generally occurs under a specific wind direction [31,32,51]. To establish a classification model that predicts VIV occurrence using structural damping ratios and wind characteristics, a decision tree is employed. The decision tree starts from a root node, and then many child nodes representing different input parameters gradually grow, forming a tree structure. Branches in the decision tree represent possible alternatives related to the conclusion. Fig. 10 shows the employed decision tree in this study to predict VIV occurrence. As shown, four branches (i.e., Branch 1, Branch 2, Branch 3 and Branch 4) are included. These four branches represent all possible events. Each branch has its outcome and occurrence probability. The adopted decision tree in Fig. 10 considers the effects of structural property (i.e., structural damping ratio) and wind characteristics (i.e., wind direction and wind speed) on VIV occurrence. The uncertainties associated with the structural property and wind characteristics are considered in the prediction of VIV occurrence. This decision tree begins with a root node (i.e., denoted as a gray node) associated with the structural damping ratio. There are two mutually exclusive events: structural damping for VIV occurrence and without the occurring of VIV (see Branch 4). The associated probabilities are P_{dar} and 1- P_{dar}, respectively. Under the premise that the structural damping is probable for VIV occurrence, the wind direction is the following parameter to identify VIV. If the wind direction is not located in the specific direction corresponding to VIV occurrence, the VIV will not occur (see Branch 3 in Fig. 10). If the wind direction corresponds to the VIV occurrence direction, the wind speed will follow to be considered for VIV occurrence prediction (see Branches 1 and 2). Branch 2 in Fig. 10 represents that the wind speed is not in the lock-in wind speed intervals for VIV. Thus, VIV does not occur. Consequently, only Branch 1

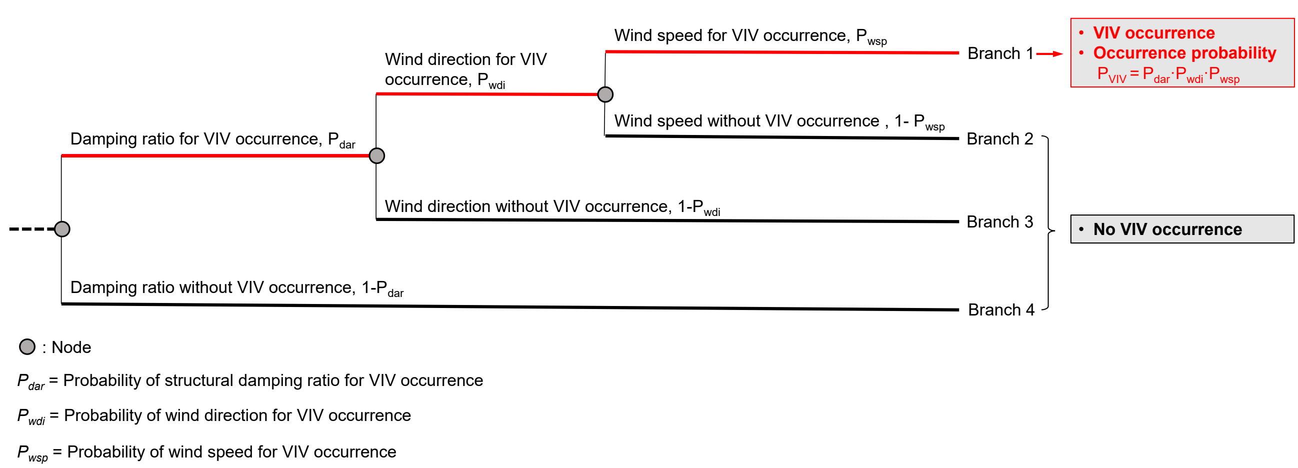

Fig. 10. Formulation of decision tree to predict VIV occurrence.

in Fig. 10 represents the event of VIV, where the wind speed remains in the lock-in wind speed intervals of VIV. Based on the assumption that the structural damping ratio, wind direction and wind speed for VIV occurrence are statistically independent events, the probability of VIV occurrence, given as P_{VIV} (see Branch 1 in Fig. 10), can be expressed as.

$$P_{VIV} = P_{dar} \cdot P_{wdi} \cdot P_{wsp} \tag{2}$$

where P_{dar}, P_{wdi} and P_{wsp} denote the probabilities of structural damping ratio, wind direction and wind speed for VIV occurrence, respectively. The decision tree model for the probabilistic VIV occurrence prediction can be validated for a few VIV events measured by the monitoring system on the bridge. Accordingly, it is determined that VIV may occur on the bridge site. Then, the management department can take appropriate and timely countermeasures. The random variables addressing the uncertainties associated with damping ratio and wind speed are used to predict P_{VIV}. The detailed computation processes of P_{dar}, P_{wdi} and P_{wsp} are described in the following sections.

5.1. Probability of structural damping ratio for VIV occurrence

With the increase of structural damping ratio in Cases 1 and 2, the VIV responses decrease. When the structural damping ratio is larger than a certain level, the VIV almost disappears or is within an acceptable level. Therefore, the damping ratio associated with the VIV occurrence has a certain range. In this study, P_{dar} calculates the probability that the structural damping belongs to that range. The uncertainty of the structural damping ratio is addressed by using a random variable with a lognormal probability density function (PDF) with a mean value and coefficient of variation (COV) of 0.50% and 0.40, respectively [4,21,22,27]. Using the PDF of structural damping ratio ξ shown in Fig. 11, the probability of structural damping ratio for VIV occurrence P_{dar} can be computed as.

$$P_{dar} = P(\zeta \leqslant \zeta_{up}) \tag{3}$$

where ξ_{up} represents the upper bound of damping ratio range for VIV occurrence. In this study, the ξ_{up} of Case 1 and Case 2 (denoted as $\xi_{up,1}$ and $\xi_{up,2}$, respectively) is 1.80% and 1.20% (see Fig. 11), respectively, as described in the previous section. As defined in Eq. (3) and Fig. 11, the probability of structural damping ratio for VIV occurrence P_{dar} is the probability that the structural damping ratio ξ is less than ξ_{up}. The area under the PDF of ξ before the ξ_{up} indicates the P_{dar}. Hence, the P_{dar} for Case 1 and Case 2 (denoted as $P_{dar,1}$ and $P_{dar,2}$, respectively) are computed as 0.999 and 0.994, respectively (see Fig. 11). It can be seen that $P_{dar,1}$ is larger than $P_{dar,2}$, and both of these values approach one, which means the increase of the structural damping ratio cannot significantly mitigate the VIV for the investigated bridge girder. Therefore, other ACMs should be considered in order to effectively decrease the probability of VIV occurrence. Additionally, the result of P_{dar} highly depends on the distribution characteristic of damping ratio. The accuracy of P_{dar} can be further improved if reliable methods for quantifying the uncertainty of damping ratios can be implemented.

5.2. Probability of wind direction for VIV occurrence

Generally, bridge VIV occurs with a specific wind direction [31–33,51]. Fig. 12 shows the schematic of the wind direction for VIV occurrence. As shown in this Figure, the wind direction of the bridge's VIV events (denoted as blue points) corresponds to the direction nearly perpendicular to the longitudinal bridge axis (denoted as red line). Therefore, to predict VIV occurrence, the effects of wind direction need to be involved. In this study, the probability of wind direction for VIV occurrence P_{wdi} is defined as.

$$P_{wdi} = \frac{N_{re,VIV}}{\sum_{i=1}^{Nwdi} N_{re,i}} \tag{4}$$

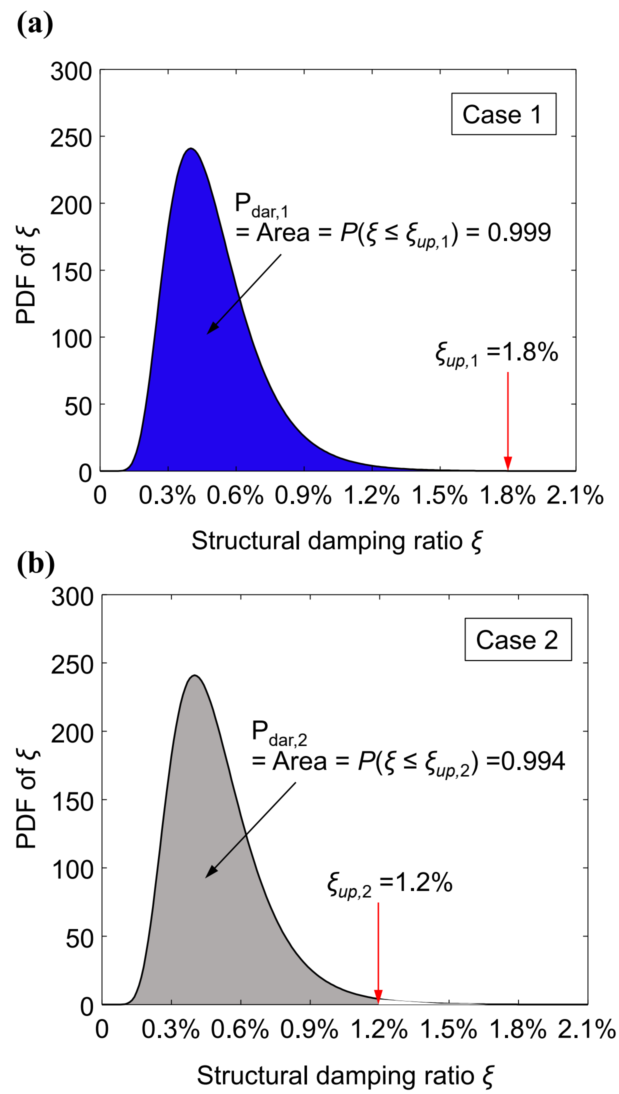

Fig. 11. PDFs of structural damping ratio for the formulation of VIV occurrence. (a) Case 1; (b) Case 2.

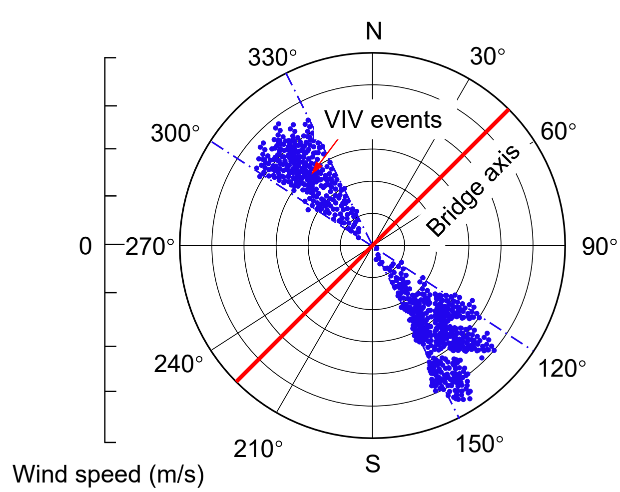

Fig. 12. Schematic of the wind direction for VIV occurrence (adapted from Xu et al. [51].

where $N_{re,VIV}$ = the number of recorded samples under the wind direction of VIV occurrence; N_{wdi} = the number of wind directions; $N_{re,i}$ = the number of recorded samples under the ith wind direction. As shown in Eq. (4), The probability of wind direction for VIV occurrence P_{wdi} is the wind-direction frequency corresponding to that of VIV occurrence.

In this study, the wind meteorological data collected through the weather station near the bridge site is used to estimate P_{wdi}. Table 3 summarizes the number of recorded samples in each wind direction at the bridge site. A total of 16 wind directions (i.e., N_{wdi} = 16, such as N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, NNW) are recorded. Finally, the wind directions corresponding to the N and S (see Table 3) are determined as the VIV occurrence directions since the bridge axis is positioned in the E-W. As a result, the P_{wdi} is computed as 0.13 by using Eq. (4).

5.3. Probability of wind speed for VIV occurrence

Inevitably, VIV of long-span bridges easily occurs in lock-in low-wind-speed intervals [6,52]. Due to the uncertain wind field, the probability of wind speed for VIV occurrence P_{wsp} needs to be considered. P_{wsp} is estimated as.

$$P_{wsp} = P(U_a \in U_{int}) \tag{5}$$

where U_a = average wind speed at the bridge site; U_{int} = lock-in wind speed interval of VIV. In this study, the U_{int} for Case 1 and Case 2 (indicated as $U_{int,1}$ and $U_{int,2}$, respectively) corresponds to the ranges of 6.16 m/s to 16.24 m/s and 8.96 m/s to 16.24 m/s, respectively. Fig. 13 (a) shows the PDF of average wind speed U_a at the bridge site and its best-fitted PDF (i.e., Generalized Extreme Value (GEV) PDF). The GEV PDF $f(U_a)$ is [23,37].

$$f(U_a) = \frac{1}{\sigma}\left(1 + k\frac{(U_a - \mu)}{\sigma}\right)^{-1-\frac{1}{k}} \cdot \exp\left[-\left(1 + k\frac{U_a - \mu}{\sigma}\right)^{-\frac{1}{k}}\right] \quad for \quad 1 + \frac{k(U_a - \mu)}{\sigma} > 0 \tag{6}$$

where σ = scale parameter; k = shape parameter; and μ = location parameter. The values of parameters σ, k and μ are 1.79, 0.02 and 4.87, respectively. The GEV PDF $f(U_a)$ shown in Fig. 13(a) is used to estimate the P_{wsp} in Eq. (5).

The average wind speed at the bridge site (see Fig. 13(a)) is obtained by the weather station near the bridge site. The area of the $f(U_a)$ belonging to U_{int} represents the P_{wsp}, as defined in Eq. (5). Fig. 13(b) and (c) show the formulation of the probability of wind speed for VIV occurrence in Case 1 and Case 2, respectively. As shown, the P_{wsp} for Cases 1 and 2 (denoted as $P_{wsp,1}$ and $P_{wsp,2}$, respectively) are 0.38 and 0.11, respectively. $P_{wsp,1}$ is larger than $P_{wsp,2}$.

Table 3
The number of recorded samples for wind directions at the bridge site.

Wind directions	Number of recorded samples	Wind directions	Number of recorded samples
N	2187	S	737
NNE	1637	SSW	667
NE	2518	SW	1105
ENE	1398	WSW	1002
E	1994	W	1012
ESE	2581	WNW	675
SE	2314	NW	837
SSE	1424	NNW	1407

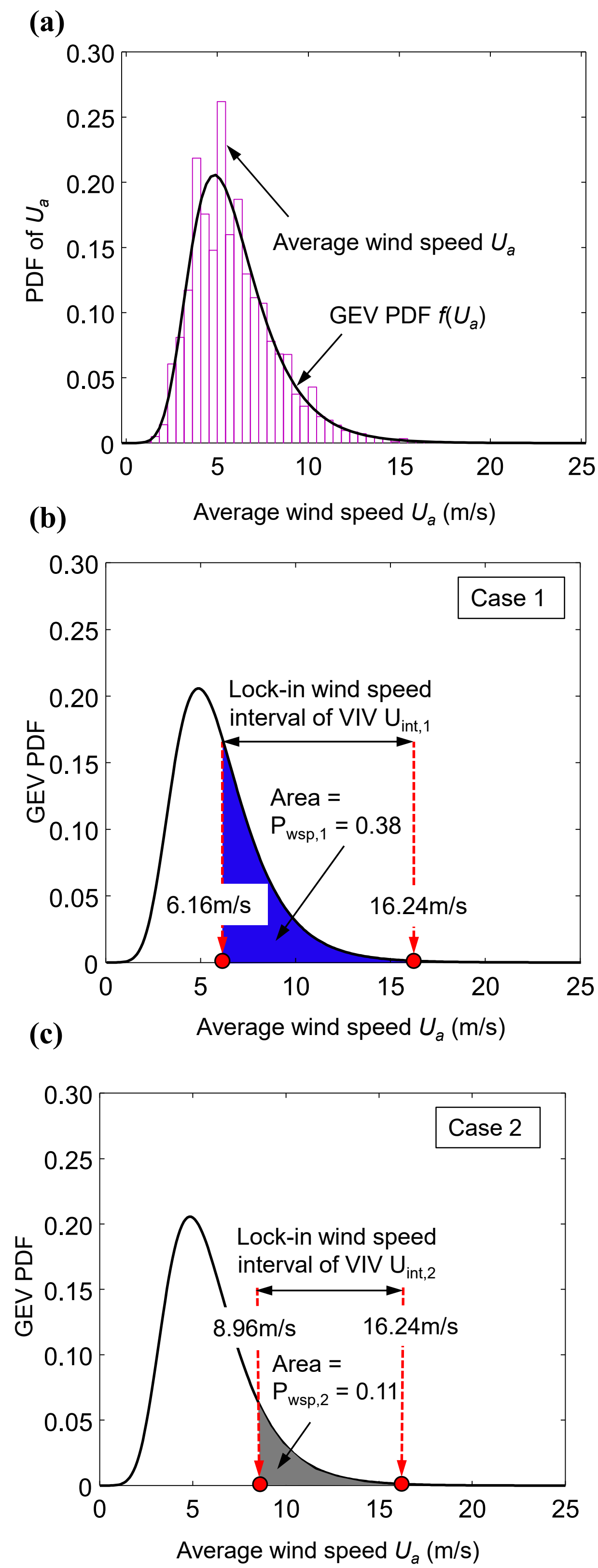

Fig. 13. (a) PDF of average wind speed; (b) formulation of the probability of wind speed for VIV occurrence in Case 1; (c) formulation of the probability of wind speed for VIV occurrence in Case 2.

5.4. Probability of VIV occurrence

The probability of VIV occurrence P_{VIV} is the product of P_{dar}, P_{wdi} and P_{wsp} as depicted in Eq. (2). The P_{dar} (Fig. 11), P_{wdi} (i.e., 0.13) and P_{wsp} (Fig. 13(b) and (c)) are used to compute the probability of VIV occurrence P_{VIV} for the investigated bridge girder. Consequently, the probabilities of VIV occurrence of this bridge girder in Case 1 and Case 2 are computed as 0.05 and 0.01, respectively. It can be seen that Case 1 results in a higher probability of VIV occurrence (i.e., 0.05) than that associated with Case 2 (i.e., 0.01). Therefore, VIV easily occurs with the low turbulence intensity. Furthermore, the probability of VIV occurrence for the investigated bridge girder is more sensitive to the changes of wind characteristics (i.e., wind speed and wind direction) than to those of the structural property (i.e., structural damping ratio) since the probability of structural damping ratio for VIV occurrence is close to one (i.e., $P_{dar,1} = 0.999$ and $P_{dar,2} = 0.994$).

6. Aerodynamic countermeasures application

VIV serving as an unexpected and unacceptable wind-induced vibration should be continuously controlled under allowable amplitudes or even extensively mitigated [25,10]. Generally, VIV countermeasures can be divided into three different categories: structural, aerodynamic and mechanical countermeasures [8,42]. ACMs are the most commonly used measures for the suppression of VIV for long-span bridges due to their practicality and simplicity. A reliable ACMs can effectively and efficiently suppress VIV by changing the bridge's configuration with extra aerodynamic components [15,26]. These aerodynamic components have been extensively studied and successfully applied to many actual bridges. Sun et al. [44] experimentally investigated the effects of handrail, inspection vehicle rail and guide vane on VIV performance of a closed box girder. The effect of installed attachments on VIV mitigation for a separated twin-box girder was proposed by Laima et al. [28]. Yang et al. [56] carried out the studies on the application of vertical stabilizer plates for suppressing VIV of a cable-supported bridge deck using wind tunnel tests.

In this study, ACMs with L-shaped windshields are used for the suppression of VIV. The effects of L-shaped windshields application on VIV performance are investigated by the wind tunnel test. Fig. 14 shows the layout of L-shaped windshields on the section model in the wind tunnel test. As illustrated in Fig. 14, the L-shaped windshields are installed close to the crash barriers on both sides of the central slot. Fig. 15 compares the vertical VIV amplitudes of the bridge girder with and without the L-shaped windshield application in Case 1. The associated vertical VIV amplitudes for the wind attack angles of $+3^\circ$, 0°, and -3° are presented in Fig. 15(a), (b) and (c), respectively. If Case 2 is tested, the bridge girder's vertical VIV amplitudes considering the installed and uninstalled windshield are obtained as shown in Fig. 16.

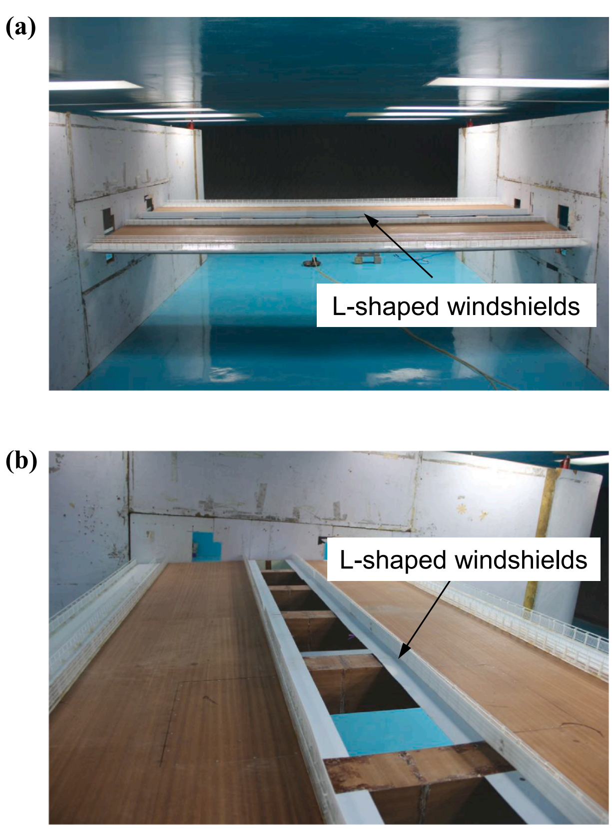

Fig. 14. Section model with L-shaped windshield in the wind tunnel test.

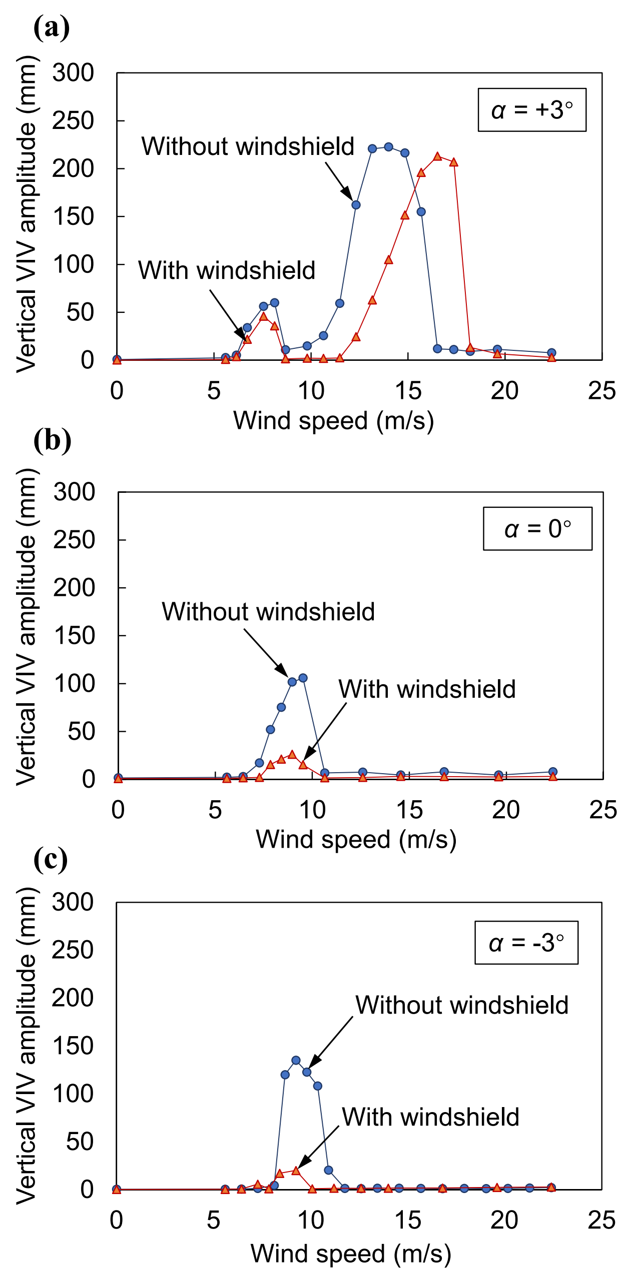

Fig. 15. Vertical VIV amplitude of the bridge girder with and without L-shaped windshield in Case 1 (a) $+3^\circ$ wind attack angle; (b) 0° wind attack angle; (a) -3° wind attack angle.

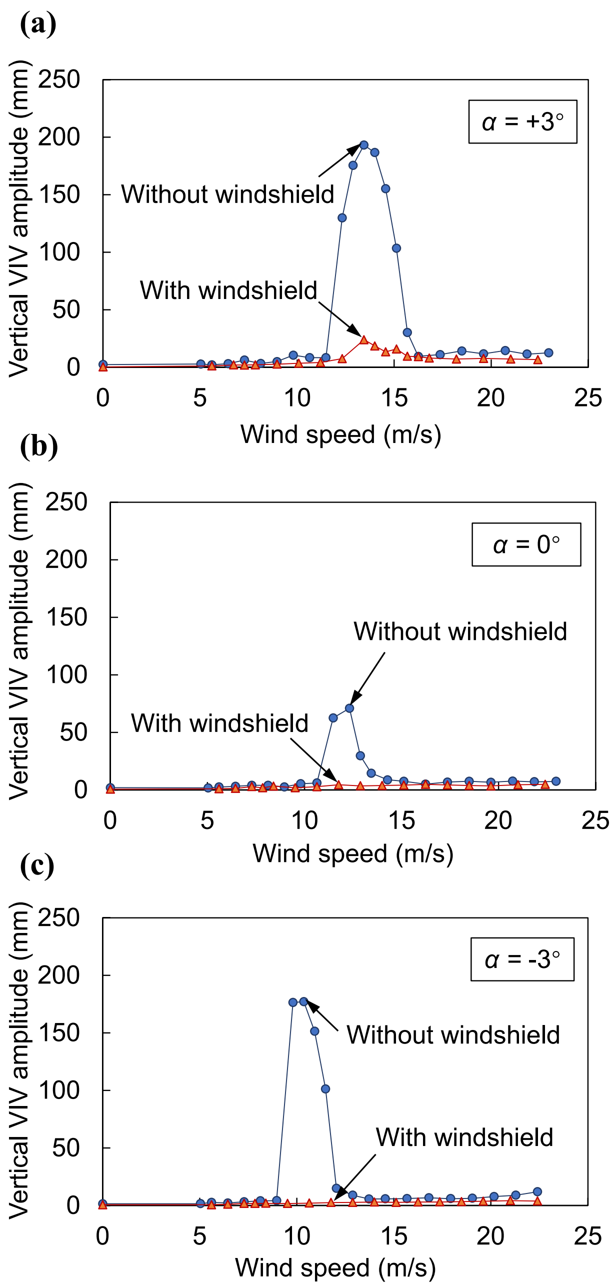

Fig. 16. Vertical VIV amplitude of the bridge girder with and without L-shaped windshield in Case 2 (a) +3° wind attack angle; (b) 0° wind attack angle; (a) −3° wind attack angle.

Furthermore, the vertical VIV amplitudes shown in Figs. 15 and 16 are based on the structural damping ratio of 0.25%. The vertical VIV amplitudes of the bridge girder with L-shaped windshields are much less than those of girders without windshield application under the wind attack angles of 0°and −3° in Case 1 (see Fig. 15(b) and (c)). VIV almost disappears under three tested wind attack angles in Case 2 when the bridge girder is installed of L-shaped windshields, as presented in Fig. 16 (a), (b) and (c). Therefore, the ACMs with L-shaped windshields can effectively suppress the VIV responses and should be further involved in the practical application.

7. Cost-benefit of aerodynamic countermeasures application

The use of ACMs with L-shaped windshields can minimize VIV responses, as described in the previous section. However, the installation of ACMs for the mitigation of VIV requires additional investment from structure managers. Cost-benefit analysis is an extensively used method to help decide whether the risk elimination measures such as ACMs need to be applied by the comparison of cost and benefit from different strategies [11,24,29]. Herein, a decision tree shown in Fig. 17 is adopted to estimate the cost-benefit from ACMs. The decision tree has two alternatives at a gray square decision node: ACMs and no ACMs application. The occurrence probability of each branch in Fig. 17 considers the probability of VIV occurrence P_{VIV} and the probability of ACMs performing their required functions P_{fun}. The probability of VIV occurrence P_{VIV} is predicted as prementioned. The probability of ACMs performing the required functions P_{fun} relies on the ACMs production's quality. There are three branches (i.e., B1, B2 and B3) associated with the alternative for the ACMs application. Branch B1 in Fig. 17 represents the event that VIV occurs and ACMs work well. The associated total cost is $C_{am} + C^1_{dir} + C^1_{indir}$ (see Branch B1 in Fig. 17), where C_{am} is the cost of ACMs application, C^1_{dir} is the direct cost of VIV including the required inspections and maintenances, and C^1_{indir} is the indirect cost caused by the traffic delay and additional vehicle fuel consumption after VIV. If ACMs do not function well during a VIV as illustrated in Branch B2, the corresponding occurrence probability and required total cost are $P_{VIV}\cdot(1 - P_{fun})$ and $C_{am} + C^0_{dir} + C^0_{indir}$, respectively. The superscripts '1' and '0' represent the cost affected by ACMs and no ACMs application, respectively. Consequently, the total monetary value for the alternative of ACMs application C_{ACM} considering the Branches B1 B2 and B3 in Fig. 17 is expressed as.

$$C_{ACM} = P_{VIV}P_{fun}\left(C_{am} + C^1_{dir} + C^1_{indir}\right) + \left[P_{VIV}\left(1 - P_{fun}\right)\right]\left(C_{am} + C^0_{dir} + C^0_{indir}\right) + \left(1 - P_{VIV}\right)C_{am} \tag{7}$$

In the same manner, the total monetary value for no ACMs C_{NACM} can be computed as.

$$C_{NACM} = P_{VIV}\left(C^0_{dir} + C^0_{indir}\right) \tag{8}$$

The detailed formulations of C_{NACM} can be found from Branches B4 and B5 in Fig. 17.

The cost-benefit of ACMs $C_{be,ACM}$, which is defined as the difference between the total monetary values C_{ACM} for ACMs and no ACMs application C_{NACM} can be expressed as follows:

$$C_{be,ACM} = C_{NACM} - C_{ACM} \tag{9}$$

Considering the uncertainties associated with C_{am}, C^1_{dir}, C^1_{indir}, C^0_{dir} and C^0_{indir}, the probability that the total monetary value for no ACMs application C_{NACM} is larger than that of ACMs C_{ACM}, which represents the probability of the positive cost-benefit from ACMs application P_{be}, can be estimated as.

$$P_{be} = P(C_{be,ACM} \geqslant 0) \tag{10}$$

Although detailed cost information on VIV risk suppression measures is difficult to obtain, some crude assumptions of design/installation costs associated with hazard mitigation strategies were suggested in the literature. For example, Lee et al. [29] proposed a probabilistic cost-benefit analysis for the seismic risk reduction technology based on a yardstick for the justifiable cost of seismic isolation system which is about 2% of the construction cost. Moreover, the inspection and repair costs were set approximately identical to the seismic isolation system installation cost. Accordingly, to assess the cost-benefit of ACMs for the investigated bridge, the cost of ACMs installation C_{am} is assumed as a unit cost lognormally distributed with the mean of 1.0 and standard deviation of 0.2 (denoted as LN(1.0; 0.2)). The effect of the indirect cost due to the traffic delay and additional vehicle fuel consumption after VIV for no ACMs application C^0_{indir} on the cost-benefit is investigated by varying the ratio *Rio* of the C_{am} to C^0_{indir}. If applied ACMs for this bridge are well-functioned during the VIV, the direct cost of VIV C^1_{dir} associated with the inspection and repairing for the structural performance

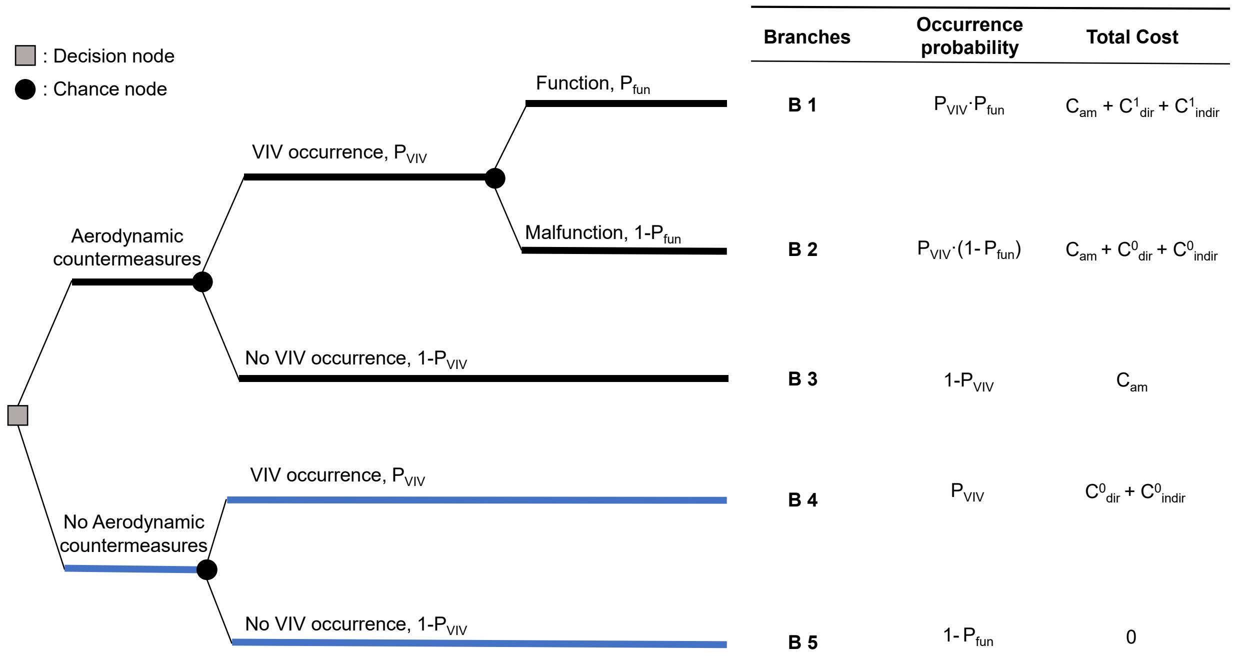

Fig. 17. Decision tree for the cost-benefit analysis of aerodynamic countermeasures.

improvement is approximately identical to the C_{am}, and there will be no indirect cost of VIV C^1_{indir} because of no traffic disruption and delay after VIV (i.e., $C^1_{indir} = 0$.). Furthermore, the ratio of the C^1_{dir} to the direct cost of VIV C^0_{dir} for no ACMs application is assumed to be 0.2. As a result, the C^0_{dir} is estimated as LN(5.0; 1.0). Monte Carlo simulation with 100,000 simples is used to formulate the C_{am}, C^1_{dir}, C^1_{indir}, C^0_{dir} and C^0_{indir}. Finally, the cost-benefit of ACMs $C_{be,ACM}$ can be estimated by Eq.(9).

Fig. 18 shows the PDFs of the cost-benefit $C_{be,ACM}$ for the probability of ACMs performing the required functions P_{fun} with 0.9 considering multiple ratios *Rio* between C_{am} and C^0_{indir} in Case 1 and Case 2. The probabilities of VIV occurrence P_{VIV} obtained as 0.05 and 0.01 for Cases 1 and 2, respectively, are used to estimate the C_{ACM} and C_{NACM} in Eq. (9). The probability that the ACMs application produces a positive cost-benefit P_{be} corresponds to the area of the PDF of $C_{be,ACM}$ above zero, as defined in Eq. (10). According to Fig. 18, it can be seen that the P_{be} increases as decreasing *Rio* (or increasing C^0_{indir}) in Case 1 and Case 2. Table 4 provides the P_{be} considering the different P_{fun} in Cases 1 and 2. As shown in Table 4, higher P_{fun} results in the higher P_{be} for a given *Rio* in two cases. Moreover, based on the same P_{fun}, Case 1 leads to the larger P_{be} than the one of Case 2 under a specific *Rio* since the probability of VIV occurrence in Case 1 (i.e., 0.05) exceeds that in Case 2 (i.e., 0.01). Therefore, compared with Case 2, ACMs application in Case 1 will be more effective when VIV occurs for the bridge.

8. Discussion on limitations of results

The example presented in this study shows the promising application of the proposed approach to predict the VIV occurrence under uncertainty for the separated twin-box girder of a long-span cable-stayed bridge. However, there are several limitations to be recognized.

The accuracy of the results presented in this study depends on the wind tunnel test and the construction of the decision tree. In the illustrated example, the effects of structural damping ratios, wind speed and wind directions on VIV performance are considered. Actually, bridge VIV involves high uncertainty resulting from wind characteristics, structural properties, and bluff-body aerodynamics. However, there is not enough information to quantify the effects of all related parameters on VIV under uncertainty. Therefore, more research efforts are needed to accurately and realistically predict the VIV occurrence under uncertainty and generalize the proposed methods by considering the various important parameters and useful information.

Although the proposed approach has the capability of predicting the VIV occurrence under uncertainty, the parameter identification associated with VIV occurrence should be further developed. For example, Yang et al. [55] systematically studied the flutter characteristics of twin-box girder bridges with the combination schemes of various slot width ratios and vertical central stabilizers through experimental investigations in conjunction with theoretical analysis. An integrated FE model to investigate the nonlinear flutter characteristics of twin-box girder suspension bridges with various slot width ratios and two wind fairing shapes was presented in Zhou et al. [60]. Parameter identification of the nonlinear aerodynamic force model for the girder with various upward vertical central stabilizers for controlling the flutter subjected to various turbulence flows was developed in Zhou et al. [61]. Accordingly, further investigations of generation mechanisms of VIV considering various parameters are required in order to fundamentally understand and accurately predict the VIV.

The probabilistic cost-benefit formulation presented in this study considers the VIV occurrence probability, costs associated with monetary loss due to the decreased structural performance by VIV and the probability of ACMs performing the required function. However, the cost of ACMs installation C_{am} is simplified as the unit cost, and other costs are estimated based on C_{am}. Also, their uncertainties are addressed by using random variables. For more practical use of the cost-benefit analysis, rational cost estimation needs to be further developed, which includes quantifying the loss from VIVs and the effect of ACMs quality on the VIV suppression [17,41,64].

9. Conclusions

A novel probabilistic approach to predict VIV occurrence under uncertainty for the separated twin-box girder of a long-span cable-stayed bridge was proposed in this paper. The wind tunnel test was conducted to identify the affected parameters for VIV occurrence. The uniform and turbulent wind flow fields under three different wind attack angles are tested. A decision tree considering the parameters of structural damping ratio, wind direction and wind speed was formulated to predict the VIV occurrence. Furthermore, the application of ACMs with L-shaped

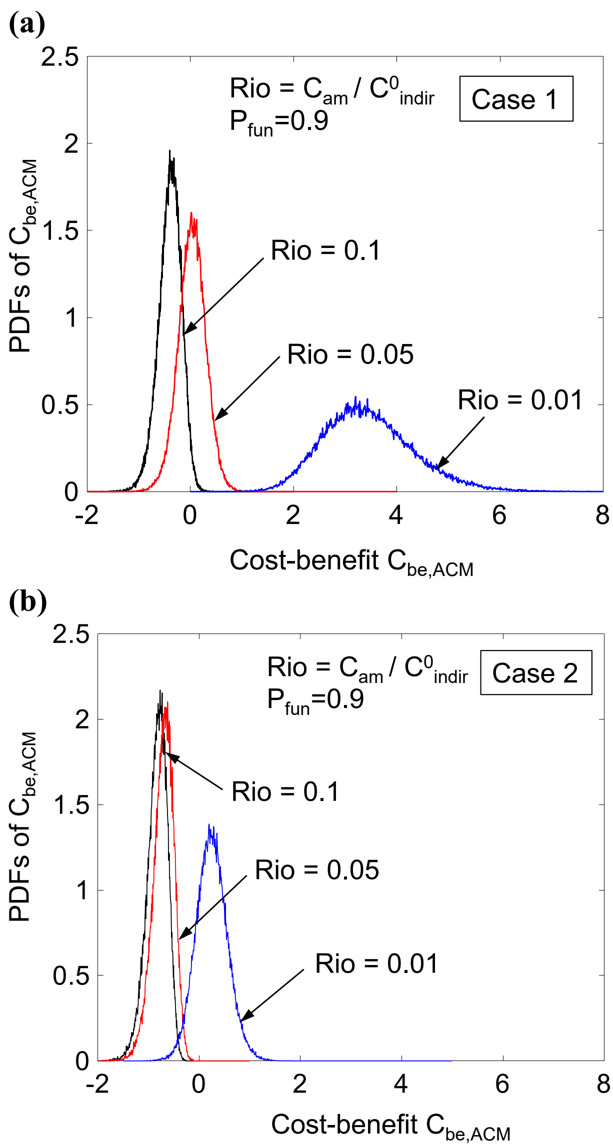

Fig. 18. PDFs of cost-benefit $C_{be,ACM}$ for $P_{fun} = 0.9$ considering different *Rio*: (a) Case 1; (b) Case 2.

Table 4
Probability of the positive cost-benefit from ACMs application for the different P_{fun} in Cases 1 and 2.

P_{fun}	Cases	*Rio*		
		Rio = 0.1	*Rio* = 0.05	*Rio* = 0.01
$P_{fun} = 0.9$	Case 1	0.0225	0.5510	0.9999
	Case 2	1.0×10^{-5}	2.0×10^{-5}	0.7897
$P_{fun} = 1.0$	Case 1	0.0585	0.7016	0.9999
	Case 2	0.0001	0.0002	0.8826

windshields was presented, taking into account the VIV performance estimation of this bridge girder and the cost-benefit analysis of ACMs. The probability that ACMs application will result in the positive cost-benefit was estimated. The conclusions can be obtained as follows:

In this study, structural damping ratio, wind speed, and wind direction are determined as the parameters for predicting VIV. With the structural damping ratio increasing, the vertical and maximum vertical VIV amplitudes decrease. Furthermore, VIV is only excited in the lock-in low-wind-speed ranges when the wind direction is nearly perpendicular to the bridge axis [20,32].

The decision tree is adopted to predict the probability of VIV occurrence, considering the probabilities of structural damping ratio, wind direction and wind speed for VIV occurrence independently. The uncertainties associated with the structural damping ratios and wind speed are addressed to predict the probability of VIV occurrence by using appropriate random variables and best-fitted distributions. For more practical use of the proposed decision tree to predict the probability of VIV occurrence, it is suggested that (a) VIV performance estimation under uncertain turbulence parameters of wind field (i.e., turbulence intensity and gust factor) (b) the effects of two important parameters (i.e., Strouhal number and Scruton number) (c) torsional VIV responses (d) correlation among the affected parameters and (e) the data from SHM systems should be further integrated.

VIV easily occurs in the low wind turbulence field since the probability of VIV occurrence in the uniform wind flow field is higher than that in the turbulent wind flow field. By comparing the effect of the structural damping ratio, the wind characteristics (i.e., wind direction and wind speed) will produce more significant influences on VIV occurrence. Therefore, effective and efficient VIV mitigation measures for decreasing the effects of wind parameters on VIV need to be further developed to guarantee structural security and reliability.

When the indirect cost of VIV for no ACMs application increases, the probability of positive cost-benefit from ACMs increases. A higher probability of ACMs performing the required function leads to a higher probability that ACMs result in a positive cost-benefit. The application of ACMs in Case 1 leads to a larger probability of positive cost-benefit compared with that in Case 2 because of the higher probability of VIV occurrence in Case 1. Therefore, the application of ACMs will be extremely effective for the bridges under VIV with a high probability.

In this study, the effects of the indirect cost of VIV for no ACMs application and the probability of ACMs performing the required functions on the probability of positive cost-benefit from ACMs are discussed. To the practical application of the proposed cost-benefit analysis, reasonable cost estimation quantifying the direct and indirect loss due to VIV should be further integrated and investigated [30,39,47].

As the damping ratio increases, the vertical VIV response decreases but cannot be significantly mitigated. Therefore, more effective mitigation measures are further required for the investigated bridge. The damping sensibility study of VIV in this study may provide a reference for wind resistance design of separated twin-box girders. The VIV of bridge girders could be significantly affected by the turbulence intensity. To be prudent, turbulence intensity of the wind field at the bridge site should be measured and analyzed carefully by bridge designers to make reasonable decisions. Although the ACM effectively suppresses the VIV, its application may not be beneficial when little VIV occurs at the bridge site. Therefore, implementing cost-benefit analyses and VIV prediction can help bridge managers make reasonable and sustainable decisions and ultimately achieve cost-effective bridge management.

CRediT authorship contribution statement

Baixue Ge: Conceptualization, Methodology, Software, Writing – original draft. **Rujin Ma:** Conceptualization, Investigation. **Fangkuan Li:** Investigation, Validation. **Xiaohong Hu:** Validation. **Airong Chen:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The support by grants from (a) Academician Project Foundation of

CCCC Award YSZX-01-2020-01-B and (b) Sponsored by Shanghai Pujiang Program Award 21PJD077 (c) National Natural Science Foundation of China Award 52178153 are gratefully acknowledged. The opinions presented in this paper are those of the authors and do not necessarily reflect the views of the sponsoring organizations.

References

[1] Bi JH, Qiao HY, Nikitas N, Guan J, Wang J, Lu P. Numerical modelling for rain wind induced vibration of cables with longitudinal ribs. J Wind Eng Ind Aerodyn 2018;178:69–79.
[2] Cao S, Zhang Y, Tian H, Ma R, Chang W, Chen A. Drive comfort and safety evaluation for vortex-induced vibration of a suspension bridge based on monitoring data. J Wind Eng Ind Aerodyn, Elsevier 2020;204:104266. https://doi.org/10.1016/j.jweia.2020.104266.
[3] Chen F, Peng H, Chan P-W, Zeng X. Wind tunnel testing of the effect of terrain on the wind characteristics of airport glide paths. J Wind Eng Ind Aerodyn, Elsevier 2020;203:104253. https://doi.org/10.1016/j.jweia.2020.104253.
[4] Davenport AG, Hill-Carroll P. Damping in tall buildings: its variability and treatment in design. In: Proceedings of a Session in Conjunction with the ASCE Convention, Seattle, WA, USA; 1986.
[5] Daniels SJ, Castro IP, Xie Z-T. Numerical analysis of freestream turbulence effects on the vortex-induced vibrations of a rectangular cylinder. J Wind Eng Ind Aerodyn, Elsevier 2016;153:13–25.
[6] Dai J, Xu Z-D, Yin X-J, Gai P-P, Luo Y. Parameters design of TMD mitigating vortex-induced vibration of the Hong Kong–Zhuhai–Macao Bridge deep-water nonnavigable bridge. J Bridge Eng 2019;24(8):06019005. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001450.
[7] Dai J, Xu Z-D, Gai P-P. Parameter determination of the tuned mass damper mitigating the vortex-induced vibration in bridges. Eng Struct, Elsevier 2020;221:111084. https://doi.org/10.1016/j.engstruct.2020.111084.
[8] Dai J, Xu Z-D, Gai P-P, Xu Y-W. Mitigation of vortex-induced vibration in bridges using semiactive tuned mass dampers. J Bridge Eng, ASCE 2021;26(6):05021003. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001719.
[9] Deng HZ, Jiang Q, Li F, Wu Y. Vortex-induced vibration tests of circular cylinders connected with typical joints in transmission towers. J Wind Eng Ind Aerodyn, Elsevier 2011;99(10):1069–78.
[10] Di F, Sun L, Chen L. Suppression of vortex-induced high-mode vibrations of a cable-damper system by an additional damper. Eng Struct, Elsevier 2021;242:112495. https://doi.org/10.1016/j.engstruct.2021.112495.
[11] Dong Y, Frangopol DM. Probabilistic life-cycle cost-benefit analysis of portfolios of buildings under flood hazard. Eng Struct, Elsevier 2017;142:290–9.
[12] El-Gammal M, Hangan H, King P. Control of vortex shedding-induced effects in a sectional bridge model by spanwise perturbation method. J Wind Eng Ind Aerodyn 2007;95(8):663–78.
[13] Frandsen JB. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. J Wind Eng Ind Aerodyn, Elsevier 2001;89(1):95–129.
[14] Fujino Y, Yoshida Y. Wind-induced vibration and control of Trans-Tokyo Bay crossing bridge. J Struct Eng, ASCE 2002;128(8);1012–25.
[15] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. Struct Eng Int, Taylor & Francis 2013;23(3):248–68.
[16] Gao D, Deng Z, Yang W, Chen W. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: A vortex-dynamics approach. J Fluids Struct, Elsevier 2021;105:103348. https://doi.org/10.1016/j.jfluidstructs.2021.103348.
[17] Goda K, Lee CS, Hong HP. Lifecycle cost–benefit analysis of isolated buildings. Struct Saf 2010;32(1):52–63.
[18] Hu C, Zhao L, Ge Y. Time-frequency evolutionary characteristics of aerodynamic forces around a streamlined closed-box girder during vortex-induced vibration. J Wind Eng Ind Aerodyn, Elsevier 2018;182:330–43.
[19] Hu C, Zhao L, Ge Y. Mechanism of suppression of vortex-induced vibrations of a streamlined closed-box girder using additional small-scale components. J Wind Eng Ind Aerodyn, Elsevier 2019;189:314–31.
[20] Huang Z, Li Y, Hua X, Chen Z, Wen Q. Automatic identification of bridge vortex-induced vibration using random decrement method. Appl Sci, MDPI 2019;9(10):2049.
[21] Kareem A. Aerodynamic response of structures with parametric uncertainties. Struct Saf, Elsevier 1988;5(3):205–25.
[22] Kareem A. Effect of parametric uncertainties on wind excited structural response. J Wind Eng Ind Aerodyn, Elsevier 1988;30(1-3):233–41.
[23] Kim S, Frangopol DM. Optimum inspection planning for minimizing fatigue damage detection delay of ship hull structures. Int J Fatigue, Elsevier 2011;33(3):448–59.
[24] Kim S, Ge B, Frangopol DM. Optimum target reliability determination for efficient service life management of bridge networks. J Bridge Eng, ASCE 2020;25(10):04020087. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001623.
[25] Kim S, Park J, Kim H. Damping identification and serviceability assessment of a cable-stayed bridge based on operational monitoring data. J Bridge Eng, ASCE, 04016123.
[26] Kwok KCS, Qin XR, Fok CH, Hitchcock PA. Wind-induced pressures around a sectional twin-deck bridge model: effects of gap-width on the aerodynamic forces and vortex shedding mechanisms. J Wind Eng Ind Aerodyn, Elsevier 2012;110:50–61.
[27] Kwon DK, Kareem A, Stansel R, Ellingwood BR. Wind load factors for dynamically sensitive structures with uncertainties. Eng Struct, Elsevier 2015;103:53–62.
[28] Laima S, Li H, Chen W, Ou J. Effects of attachments on aerodynamic characteristics and vortex-induced vibration of twin-box girder. J Fluids Struct, Elsevier 2018;77:115–33.
[29] Lee H-P, Kim S, Cho M-S, Ji Y-S. Application of sliding seismic isolator to building structures considering cost, performance and inspection: A case study. Struct Infrastruct Eng, ASCE 2015;11(7):851–68.
[30] Le V, Caracoglia L. Life-cycle cost assessment of vertical structures under nonstationary winds: Downburst vs. tornado loads. Eng Struct, Elsevier 2021;243:112515. https://doi.org/10.1016/j.engstruct.2021.112515.
[31] Li H, Laima S, Ou J, Zhao X, Zhou W, Yu Y, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. Eng Struct, Elsevier 2011;33(6):1894–907.
[32] Li H, Laima S, Zhang Q, Li Na, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. J Wind Eng Ind Aerodyn, Elsevier 2014;124:54–67.
[33] Li S, Laima S, Li H. Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression. J Wind Eng Ind Aerodynam, Elsevier 2018;172:196–211.
[34] Li L, Wu T, He X, Hao J, Wang H, Xu H. Reliability evaluation of vortex-induced vibration for a long-span arch bridge. J Bridge Eng, ASCE 2018;23(5):05018002. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001225.
[35] Li M, Sun Y, Jing H, Li M. Vortex-induced vibration optimization of a wide streamline box girder by wind tunnel test. KSCE J Civ Eng, Springer 2018;22(12):5143–53.
[36] Liu Z, Shen J, Li S, Chen Z, Ou Q, Xin D. Experimental study on high-mode vortex-induced vibration of stay cable and its aerodynamic countermeasures. J Fluids Struct, Elsevier 2021;100:103195. https://doi.org/10.1016/j.jfluidstructs.2020.103195.
[37] Martins ES, Stedinger JR. Generalized maximum-likelihood generalized extreme-value quantile estimators for hydrologic data. Water Resour Res, AGU 2000;36(3):737–44.
[38] Ministry of Transport of the People's Republic of China. Wind-resistant Design Specification for Highway Bridges (JTG/T 3360-01-2018), China Communications Press, Beijing, China; 2018.
[39] Micheli L, Alipour A, Laflamme S, Sarkar P. Performance-based design with life-cycle cost assessment for damping systems integrated in wind excited tall buildings. Eng Struct, Elsevier 2019;195:438–51.
[40] Owen JS, Vann AM, Davies JP, Blakeborough A. The prototype testing of Kessock Bridge: response to vortex shedding. J Wind Eng Ind Aerodyn, Elsevier 1996;60:91–108.
[41] Padgett JE, Dennemann K, Ghosh J. Risk-based seismic life-cycle cost–benefit (LCC-B) analysis for bridge retrofit assessment. Struct Saf 2010;32(3):165–73.
[42] Sarwar MW, Ishihara T. Numerical study on suppression of vortex-induced vibrations of box girder bridge section by aerodynamic countermeasures. J Wind Eng Ind Aerodyn, Elsevier 2010;98(12):701–11.
[43] Shiraishi N, Matsumoto M. On classification of vortex-induced oscillation and its application for bridge structure. J Wind Eng Ind Aerodyn, Elsevier 1983;14:419–30.
[44] Sun Y, Liao H, Li M. Mitigation measures of vortex-induced vibration of suspension bridge based on section model test. J Southwest Jiaotong Univ 2012;47(2):218–23. in Chinese.
[45] Sun Z, Zou Z, Ying X, Li X. Tuned mass dampers for wind-induced vibration control of chongqi bridge. J Bridge Eng 2020;25(1):05019014. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001510.
[46] Trush A, Pospíšil S, Kuznetsov S, Kozmar H. Wind-tunnel experiments on vortex-induced vibration of rough bridge cables. *J Bridge Eng*, ASCE 2017;22(10):06017001. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001104.
[47] Wang CS, Zhai MS, Li HT, Ni YQ, Guo T. Life-cycle cost based maintenance and rehabilitation strategies for cable supported bridges. Int J Adv Steel Construct 2015;11(3):395–410.
[48] Wang C, Hua X, Feng Z, Xiao H, Chen Z, Zhang J. Experimental investigation on vortex-induced vibrations of a triple-box girder with web modification. J Wind Eng Ind Aerodyn 2021;218:104783. https://doi.org/10.1016/j.jweia.2021.104783.
[49] Williamson CHK, Govardhan R. A brief review of recent results in vortex-induced vibrations. *J Wind Eng Ind Aerodyn*, Elsevier 2008;96(6-7):713–35.
[50] Wu T, Kareem A. Vortex-induced vibration of bridge decks: Volterra series-based model. *J Eng Mech*, ASCE 2013;139(12):1831–43.
[51] Xu S, Ma R, Wang D, Chen A, Tian H. Prediction analysis of vortex-induced vibration of long-span suspension bridge based on monitoring data. *J Wind Eng Ind Aerodyn*, Elsevier 2019;191:312–24.
[52] Xu K, Ge Y, Zhao L. Quantitative evaluation of empirical models of vortex-induced vibration of bridge decks through sectional model wind tunnel testing. *Eng Struct*, Elsevier 2020;219:110860. https://doi.org/10.1016/j.engstruct.2020.110860.
[53] Yang Y, Zhou R, Ge Y, Mohotti D, Mendis P. Aerodynamic instability performance of twin box girders for long-span bridges. J Wind Eng Ind Aerodyn 2015;145:196–208.
[54] Yang Y, Zhou R, Ge Y, Zhang L. Experimental studies on VIV performance and countermeasures for twin-box girder bridges with various slot width ratios. J Fluids Struct 2016;66:476–89.
[55] Yang Y, Zhou R, Ge Y, Zou X, Zhang L. "Flutter characteristics of twin-box girder bridges with vertical central stabilizers. Eng Struct 2017;133:33–48.
[56] Yang Y, Kim S, Hwang Y, Kim HK. Experimental study on suppression of vortex-induced vibration of bridge deck using vertical stabilizer plates. *J Wind Eng Ind Aerodyn*, Elsevier 2021;210:104512.

[57] Yu H, Xu F, Ma C, Zhou A, Zhang M. Experimental research on the aerodynamic responses of a long-span pipeline suspension bridge. *Eng Struct*, Elsevier 2021;245: 112856. https://doi.org/10.1016/j.engstruct.2021.112856.

[58] Zhou L, Ge Y. Wind tunnel test for vortex-induced vibration of vehicle-bridge system section model. J Braz Soc Mech Sci Eng 2008;30(2):110–7.

[59] Zhou R, Yang Y, Ge Y, Mendis P, Mohotti D. Practical countermeasures for the aerodynamic performance of long-span cable-stayed bridges with open decks. Wind Struct 2015;21(2):223–39.

[60] Zhou R, Ge Y, Yang Y, Du Y, Zhang L. Wind-induced nonlinear behaviors of twin-box girder bridges with various aerodynamic shapes. Nonlinear Dyn 2018;94(2): 1095–115.

[61] Zhou R, Ge Y, Liu S, Yang Y, Du Y, Zhang L. Nonlinear flutter control of a long-span closed-box girder bridge with vertical stabilizers subjected to various turbulence flows. Thin-Walled Struct 2020;149:106245. https://doi.org/10.1016/j.tws.2019.106245.

[62] Zhou Q, Zheng S, Chen Z, Yang F, Liu J. Vortex-induced vibration reliability analysis in cable-stayed bridges: wind and aerodynamic uncertainties. *Struct Eng Int*, Taylor & Francis 2021;31(1):51–9.

[63] Zhou R, Ge Y, Liu Q, Yang Y, Zhang L. Experimental and numerical studies of wind-resistance performance of twin-box girder bridges with various grid plates. Thin-Walled Struct 2021;166:108088. https://doi.org/10.1016/j.tws.2021.108088.

[64] Zou G, Banisoleiman K, González A, Faber MH. Probabilistic investigations into the value of information: a comparison of condition-based and time-based maintenance strategies. Ocean Eng 2019;188:106181. https://doi.org/10.1016/j.oceaneng.2019.106181.